Feature 1: Card Counting & Risk Analysis Engine

- **Target Feature**: The detection engine module that tracks the running/true count based exclusively on public CardObservation records and evaluates whether a player's betting and strategy patterns repeatedly correlate with the count across multiple rounds.
- **Failure Modes Covered**: Incorrect arithmetic tracking of high/low cards, failure to reset counts upon a shuffle event, unhandled missing sensor data, and false positives generated by normal winning streaks.

1. Unit Test Set (Feature 1)

- **Target Component**: Round reconstruction and CardObservation public state logic.
- **Test Method / Vector**: Mock a sequence of incoming CardObservation entries with specific ranks (e.g., 2–6 valued at +1, 7–9 at 0, 10–Ace at 1) and various timestamps. Invoke isPublicAt(DateTime) to filter out non-public or hidden cards.
- **Expected Result**: The calculation module computes the exact running count using only cards exposed before the player's decision, ignoring hidden dealer cards or future shoe orders.

2. Integration Test Set (Feature 1)

- **Target Components**: Integration between Round, PlayerSession, and Hand data flow.
- **Test Method / Vector**: Simulate a multi-round session where a mock player systematically increases wager values in PlayerSession and modifies action choices in Hand whenever the true count exceeds +2.
- **Expected Result**: The system correctly links the historical betting trends to the public count timeline, passing a structured correlation matrix to the risk model without dropping data across round boundaries.

3. System Test Set (Feature 1)

- **Target Component**: End-to-end detection workflow from table data ingestion to the Alert Dashboard.
- **Test Method / Vector**: Inject a continuous live stream of 50 simulated rounds containing high card-counting correlation, interrupted midway by a recordShuffle() call.
- **Expected Result**: The system properly resets the running count upon shuffle detection, avoids guessing hidden cards, filters out false positives due to short winning streaks, and outputs a qualified alert with confidence scoring and video references within the 10-second requirement.

Feature 2: Sensor Reconciliation & Real-Time Monitoring Subsystem

- **Target Feature**: The ingestion and conflict-resolution subsystem that processes hardware inputs from CaptureDevice instances (seat sensors, dealer readers, and cameras), handles short network outages via local buffering, and updates dashboard metrics.
- **Failure Modes Covered**: Conflicting inputs between seat sensors and camera feeds, dropped data packets during network glitches, and unhandled hardware failures.

1. Unit Test Set (Feature 2)

- **Target Component**: CaptureDevice and CardObservation confidence validation.
- **Test Method / Vector**: Input conflicting sensor observations for the same seat and timestamp where a seat sensor reads an Ace of Spades with 90% confidence, but the camera feed reads a King of Spades with 60% confidence.
- **Expected Result**: The validation function successfully flags the discrepancy, applies the threshold rules, lowers the overall observation confidence, and prompts the fallback rule to reference the dealer feed.

2. Integration Test Set (Feature 2)

- **Target Components**: CaptureDevice event collector interacting with the local buffering database during network partitions.
- **Test Method / Vector**: Simulate a temporary network outage where table equipment continues generating unique ID events, followed by network restoration and database flushing.
- **Expected Result**: Buffered events with synchronized timestamps are successfully flushed, deduplicated, and processed in correct chronological order by the event collector without crashing.

3. System Test Set (Feature 2)

- **Target Component**: Dashboard UI and operator device-health monitoring.
- **Test Method / Vector**: Trigger a simulated hardware malfunction (checkHealth() returns false) on an active table seat sensor.
- **Expected Result**: The dashboard immediately displays a warning regarding the unavailable device, pauses scoring for that specific seat to prevent inaccurate counts, and allows an operator to view device status and log notes.